communications medicine

A Nature Portfolio journal

Review article

https://doi.org/10.1038/s43856-026-01823-z

Evidence, use cases, and implementation safeguards of large language models in primary care

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Recent developments in large language models (LLMs) have created new opportunities to support primary care, where much of clinical work is text-mediated. This narrative review synthesizes evidence on LLM applications relevant to primary care workflows and summarizes implementation safeguards. Across studies, the most consistently supported near-term value is workflow augmentation, particularly documentation and inbox management (e.g., drafting portal replies and summarizing information for clinician review) and communication support, where benefits are reported primarily as process endpoints (time, acceptability, perceived communication quality) rather than hard patient outcomes. Evidence for improvements in clinician diagnostic reasoning, treatment planning, and downstream patient outcomes is more limited and context-dependent, and many evaluations remain simulated or conducted in adjacent settings, limiting generalizability to routine primary care. Accordingly, potential roles in population health and cost reduction should be treated as hypothesis-generating and evaluated prospectively. Challenges related to privacy, security, transparency, and model reliability shape organizational governance requirements and evolving regulatory expectations for the clinical use of generative AI in primary care. We emphasize a pragmatic adoption approach: prioritize high-volume, lower-risk clerical and communication workflows; maintain clinician verification and accountability; and apply governance and equity safeguards (e.g. privacy, security, transparency, auditability, monitoring for drift and error) before scale-up.

Advancements in artificial intelligence (AI)[1–3], as well as in large language models (LLMs) such as GPT-3 (2020) and GPT-4 (2023), have created new avenues for automating medical administrative tasks across various specialties. LLMs represent a specific class of AI that harnesses extensive text data to produce human-like linguistic outputs, comprehend contextual nuances, and assist in diverse functions, including information summarization, communication drafting, and decision support. Because primary care relies heavily on text-mediated workflows, LLMs are most plausibly useful for communication tasks (e.g., drafting patient portal replies, visit notes, discharge summaries, referral letters, and patient education text) and for management of clinical information and workflow artifacts (e.g., summarizing histories, reconciling lists and medications), highlighting abnormal results, and routing inbox tasks for clinician review.

Concurrently, primary care environments are facing escalating challenges (e.g., increased workload due to increasing patient volumes, complex administrative demands, and a transition toward value-based care models that require extensive documentation and reporting) that adversely impact healthcare delivery and clinician well-being. As a result, clinicians often dedicate an inordinate amount of time to non-clinical tasks, such as inputting information into electronic health records (EHRs), which diminishes their capacity for patient care and contributes to clinician burnout.

The objective of this manuscript is to summarize and elaborate on existing studies evaluating the potential of LLMs to alleviate some of the significant burdens encountered by primary care physicians (PCPs) and improve practice. Specifically, it seeks to assess the capacity of LLMs to optimize administrative tasks, improve clinician-patient communication,

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and enhance overall efficiency within primary care workflows, thereby informing policy decisions. In this manuscript, "policy" refers to the health-system governance and regulatory requirements that determine when and how LLMs can be used in primary care (e.g., acceptable-use rules, oversight and accountability, documentation and auditability, privacy/security controls, and monitoring for safety and equity). Finally, by examining practical applications, feasibility, and limitations of LLM integration in primary care, this review synthesizes best practices for workflow deployment, identifies key evidence gaps (e.g., limited pragmatic evaluations, uncertain generalizability, and incomplete reporting), summarizes major implementation challenges (e.g., privacy/security, reliability, bias, and governance), and offers recommendations for clinicians, technologists, and healthcare organizations (Fig. 1; Table 1).

Following a brief description of scope, approach, and a qualitative evidence appraisal (including common limitations, such as bias, generalizability, transparency, and model drift), this narrative review synthesizes emerging evidence on LLM applications relevant to primary care workflows and translates heterogeneous study designs into practical implications for clinical adoption.

We next present primary care-relevant Endpoints and LLM Workflow Impacts, beginning with Definitions of Outcomes and Endpoints and proceeding through the included domains/subsections (Diagnostic Reasoning and Decision Support; Electronic Health Records; Diabetes; Geriatric Care; Cardiology; Radiology; Neurology; Medical Knowledge Assessment and Educational Support; Population health and prevention in primary care; and Specialist-informed Decision Support in Primary Care: Escalation and Referral).

We then address Cost-minimizing by using LLMs in Primary Care, followed by LLM Governance and Policy Considerations for Primary Care Deployment and Ethical and Equity Considerations.

We (1) organize the literature by primary-care tasks (documentation, triage/routing, patient communication, and decision support), (2) distinguish evidence generated in primary care from task-relevant evidence in adjacent settings, and (3) provide a pragmatic safeguards checklist (oversight, grounding to trusted sources, monitoring, and equity evaluation) to inform implementation and policy decisions in primary care; Fig. 1 maps use-cases and the evidence strength, while Table 1 consolidates best practices, gaps, and implementation recommendations for primary care. This framing is intended to help clinicians and health systems decide where LLMs are most likely to deliver near-term value and what must be validated before scale-up.

Scope and organization

This is a narrative review intended to synthesize representative evidence rather than provide an exhaustive inventory. For clarity, we organize the manuscript around primary-care workflows: (1) documentation and inbox management (ongoing stream of non-visit clinical work, such as patient portal messages, test results, refill and prior authorization requests, incoming consult notes and discharge summaries, care gap alerts, task routing/triage), (2) patient communication and triage/routing, (3) clinician decision support, and (4) chronic disease/risk management. Evidence from adjacent settings (e.g., emergency departments or specialty services) is included only when task-relevant and is explicitly labeled as such; in this review, "task-relevant" adjacent-setting evidence is included specifically to inform primary care referral and urgency decisions, triage and routing,

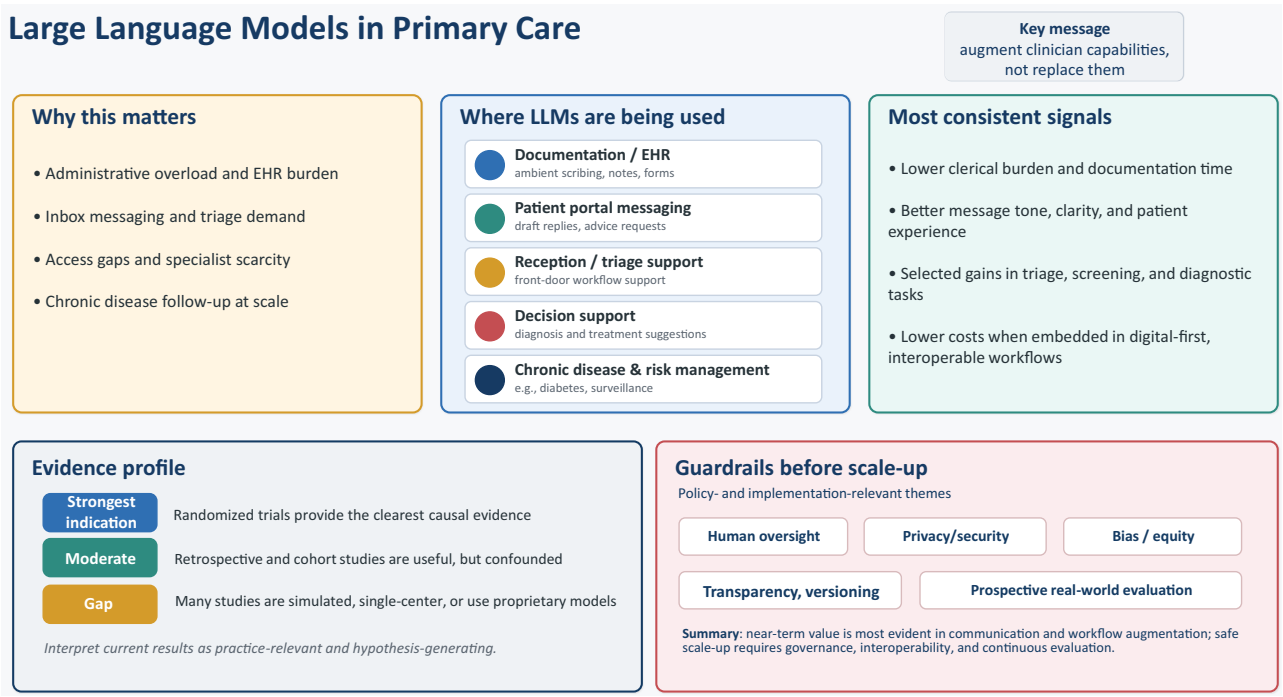


Fig. 1 | Central Illustration. Graphical overview of LLM use-cases and evidence strength in primary care–relevant workflows. This narrative review synthesizes emerging evidence on how large language models (LLMs) are being evaluated and deployed across primary care-relevant workflows. LLM use-cases cluster into (**1**) documentation and inbox management, (**2**) patient communication and triage support, (**3**) clinical reasoning and decision support, and (**4**) disease-focused applications. Across studies, reported benefits most consistently involve efficiency and communication quality, whereas improvements in diagnostic performance are variable and highly context dependent. Common evidence constraints include simulated or retrospective evaluations, limited reporting of model versioning and implementation details, and sparse real-world outcomes. We highlight pragmatic safeguards for adoption, including human oversight, privacy-aware deployment, monitoring for drift and error, and equity considerations. The "Where LLMs are being used" panel is domain-coded: colored markers identify workflow use-case categories (documentation/EHR; patient messaging; reception/triage; decision support; chronic disease/risk management). These domain colors are not an evidence rating. Evidence strength is encoded only in the "Evidence profile" panel, which uses a separate key (e.g., strongest signal = randomized trials; moderate = observational/retrospective studies; gap = simulated/single-center/proprietary evaluations). Accordingly, the red and dark markers in the use-case list denote specific domains (e.g., decision support; chronic disease/risk), not higher or lower evidence.

Table. 1 | Best practices, gaps and challenges, and recommendations in using LLMs in primary care

Best practices	Description
Integration of LLMs into workflows	Implement LLMs to automate repetitive tasks, such as triaging, summarizing patient history, and drafting message responses in electronic health records (EHRs). This streamlines clinician workflows, reduces administrative burdens, and allows more time for direct patient care[23,34]
Patient education	Utilize LLMs to convert complex medical terms into simple, patient-friendly language. By ensuring clarity and empathy in responses, LLMs improve patient comprehension and adherence to treatment plans, especially in diverse cultural and linguistic contexts[26,42]
Proactive risk management	Leverage AI to flag potential risks, such as polypharmacy issues, drug interactions, or early signs of chronic disease. In geriatric and chronic care settings, this enhances safety by proactively identifying and addressing care gaps[8,39]
Contextual decision support	Equip LLMs with retrieval-augmented generation (RAG) capabilities to provide real-time, evidence-based insights during clinical encounters. This enhances diagnostic accuracy and ensures clinical recommendations are grounded in the latest guidelines[26,41]
Gaps & challenges	Description
Accuracy and hallucinations	LLMs can produce factually incorrect outputs or "hallucinations," particularly when the input query is poorly defined or when relevant, context-specific data is unavailable. This can compromise patient safety and decision-making in critical scenarios[43,71]
Data privacy concerns	Ensuring HIPAA compliance is a significant challenge when integrating sensitive patient data with LLMs. Cloud-based AI systems are particularly vulnerable to cybersecurity risks, necessitating robust encryption, secure storage, and controlled access[55,66]
Explainability and trust	The "black-box" nature of LLMs often obscures their reasoning processes, creating mistrust among clinicians. Without clear, interpretable insights into how decisions are made, healthcare professionals may be reluctant to adopt these technologies[43,55]
Bias in decision-making	LLMs trained on biased datasets risk perpetuating inequities, disproportionately impacting vulnerable populations. For example, underrepresented demographics in training data may lead to inaccurate recommendations or incomplete care[38]
	Recommendations
Retrieval-augmented generation	Develop RAG systems that incorporate real-time, context-specific knowledge from EHRs, clinical guidelines, and recent peer-reviewed literature. This reduces hallucinations and enhances decision-making by grounding LLM outputs in validated data[41,42]
Privacy-preserving mechanisms	Implement advanced privacy-preserving techniques such as federated learning, which trains models without transferring sensitive data to central servers, or self-hosted AI systems for maximum data control and regulatory compliance[66,71]
AI alignment	Develop explainability tools to allow clinicians to understand and validate AI-generated recommendations. By incorporating transparent reasoning and visualization methods, these tools can enhance trust and facilitate informed decision-making[43,66]
Standard evaluation	Establish comprehensive evaluation metrics and frameworks tailored for healthcare. Metrics such as faithfulness, clinical relevance, and patient safety should be assessed in real-world trials before wide deployment of LLMs in primary care[38,72]
Robust dataset development	Ensure that training datasets are diverse and representative of all demographic groups. Include data on underrepresented populations to mitigate biases[6,7] and promote equitable healthcare delivery[38]

transitions of care, surveillance, and follow-up of abnormal and incidental findings, as well as specialist-informed decision support in underserved settings. We then synthesize implementation considerations, including safety, governance, ethical, legal, and equity issues, as well as health-economic implications, summarized in Fig. 1 and Table 1. We use Fig. 1 as the organizing map of primary care workflow categories and evidence strength, and Table 1 as the consolidated implementation checklist referenced throughout the sections below.

We included evidence from adjacent settings (e.g., emergency care or specialty services) only when the task structure is central to primary care (e.g., triage/routing), and we explicitly label these studies as adjacent-setting evidence rather than primary-care-setting evidence. Extracted metrics and study attributes are summarized in Online Supplement Table 1.

Search methodology

Articles for this narrative review were selected based on the role of LLMs in primary care, following two searches on PubMed; consistent with the narrative-review design, the selected PubMed searches were not intended to provide an exhaustive identification of all relevant literature. The first search employed the keywords "Primary Care" and "Large Language Models," yielding 55 results, while the second search, using "Large Language Models in Primary Care," returned 585 results. For both searches, abstracts and introductions were screened for relevance to either LLMs in healthcare and/or LLMs in primary care. Articles that explicitly mentioned both "LLMs" AND "primary care" in the article were included, resulting in the inclusion of 9 articles from the first search and 14 from the second search. If these words were not present, then the article was not included in our narrative review. Subsequently, the selected articles were categorized according to their primary focus, including applications in diabetes care, general practice, geriatric care, health records, cardiology, radiology, neurology, mental health, and ethical considerations (Online Supplementary Fig. 1).

Evidence appraisal

A qualitative appraisal of the selected literature reveals a landscape of promising but early-stage evidence, with varying levels of robustness and common methodological limitations (Online Supplement Table 1 and 2).

Higher-strength evidence. The most robust evidence comes from randomized controlled trials (RCTs), such as those by Goh et al.[4] and Wan et al.[5]. These studies support causal inferences about LLM efficacy in controlled and real-world settings, respectively, and their design reduces selection and performance biases[6,7].

Moderate-strength evidence. Several studies employed retrospective evaluations[8,9] or prospective cohort designs. While these provide valuable real-world performance data, they are susceptible to confounding factors and selection bias[6,7]. Benchmarking studies[9,10] are essential for technical validation but often use simulated vignettes or historical datasets, limiting their generalizability to dynamic clinical environments.

Feasibility and attitudes evidence. Cross-sectional surveys[11] are crucial for understanding clinician attitudes and early adoption patterns, but are subject to respondent bias (e.g., early adopters may be overrepresented) and cannot establish causality.

Common limitations across studies

Simulation vs. real-world impact. A large share of the evidence base comes from simulated cases, vignettes, or retrospective datasets rather than prospective deployment in routine primary care. As a result, improvements observed under controlled conditions may not translate directly to real-world workflow integration, safety, and patient-outcome impact, and longer-term studies are needed to understand downstream effects (including automation bias)[6,7].

Generalizability. Generalizability remains uncertain because many evaluations are confined to single institutions or specific national health systems (e.g., China, the UK, the US), where local documentation practices, language, patient mix, and regulatory context can shape performance. Broader external validation across settings, populations, and EHR ecosystems is needed before results can be assumed to hold across primary care.

Technical Transparency. Transparency is further limited by incomplete technical reporting, particularly when proprietary, rapidly changing commercial models are used. Missing details on model versioning, prompts, data provenance, fine-tuning, and evaluation protocols make it difficult to reproduce findings and to determine when, and why, performance changes or degrades.

Model performance drift. The rapid pace of LLM iteration introduces an additional limitation: technical obsolescence. Many evaluations rely on model versions that may be superseded within months, and commercial deployments may change model behavior through updates that are not fully transparent to end users[12]. As a result, performance estimates and qualitative findings from prior versions can quickly become uninformative for decisions about current-state models and workflows. To mitigate this, studies should report the model identifier and access tier, version (or release date when versioning is not disclosed), date(s) of use, and any prompt templates or system settings used; thus, health systems should treat LLM tools as "living" components requiring periodic revalidation, drift monitoring, and re-assessment of safety and equity when models, prompts, or clinical contexts change[12].

Conflict of Interest. Although many papers declare no conflicts, the field's rapid commercialization warrants careful attention to funding sources, affiliations, and reporting incentives. Clear disclosure and independent replication are especially important when studies evaluate vendor-provided tools or are conducted in product-development contexts.

Types of large language models

LLMs have revolutionized natural language processing, with two primary categories emerging:

General-purpose LLMs. such as GPT-3/4 by OpenAI and Google's BERT, are trained on vast and diverse datasets encompassing a wide range of internet text and books[13]. This extensive training allows them to exhibit remarkable versatility across numerous tasks, including text generation, summarization, translation, and question answering in various domains. For example, GPT-4 demonstrates enhanced accuracy, a larger context window, and improved understanding of nuance compared to its predecessors, even incorporating multimodal capabilities like image input[14].

Domain-specific LLMs. like Google's MedPaLM and Microsoft's BioGPT, are fine-tuned or pre-trained on specialized datasets relevant to a particular field. MedPaLM, for example, is tailored for the medical domain, trained on medical literature, and designed to excel in tasks like medical question answering, dialogue generation, and even medical image segmentation[15]. BioGPT, similarly, focuses on biomedical text generation and mining, pre-trained on millions of PubMed abstracts to provide high-quality insights for drug discovery, clinical research, and patient data interpretation[16].

While general-purpose LLMs offer broad applicability, domain-specific models provide superior accuracy and contextual understanding within their niche due to their focused use of training data. This specialization makes them valuable in fields where precision and reliability are paramount, such as healthcare or legal research, where even minor inaccuracies can have significant consequences. The choice between general-purpose and domain-specific LLMs often depends on the task's requirements, balancing versatility against the need for deep, accurate domain expertise.

Primary care relevant endpoints and LLM workflow impacts

Definitions of outcomes and endpoints. In the current evidence base, most studies involving LLMs in primary care report workflow/process endpoints (e.g., documentation time, message drafting acceptability, triage/routing performance) and intermediate clinical-decision metrics (e.g., diagnostic reasoning scores on vignettes), rather than hard patient outcomes (e.g., hospitalization, mortality). Accordingly, we use "outcomes" here to refer to the primary endpoints reported in each study, and we explicitly note where patient-level outcomes are measured or where evidence remains limited.

Diagnostic reasoning and decision support. One study that looked at diagnostic reasoning[17] within general practice showed that simply providing physicians with access to an LLM chatbot does not automatically enhance diagnostic reasoning in primary care environments, despite the LLM's ability to outperform physicians on constrained, vignette-based diagnostic tasks when provided the same case information. Here, "outperform" refers to accuracy on a closed-set reasoning task and does not imply readiness for autonomous diagnosis in real clinical encounters. The results indicated that while LLMs hold potential benefits, they required more than just access to significantly improve physician diagnostic performance[18]; effective training in prompting techniques and integration into clinical workflows are essential to fully harness their capabilities[11]. The study highlighted the crucial role of physician oversight, as LLMs are not yet equipped for independent diagnosis, particularly due to the complexities involved in patient interviews and data gathering. It also points out the need for comprehensive research and improved evaluation methods to accurately measure the impact of LLMs on clinical decision-making, especially in practical settings[17,19].

Electronic health records. Primary care clinicians (and supporting staff) spend substantial time reviewing and reconciling prior documentation and test results and then updating the EHR with visit notes, orders, referrals, and follow-up plans; this documentation and inbox management are a major source of administrative burden.

In this context, an ambient scribe is an AI-assisted documentation tool that captures the clinician-patient encounter (e.g., via visit audio) and produces a draft clinical note and related documentation (i.e., orders, updates, after-visit summaries, etc.) for clinician review[20–22]. In primary care, ambient scribes are most relevant because they can reduce time spent on visit-note documentation, enabling clinicians to focus on patient interaction and on higher-value clinical tasks.

LLMs have the potential to alleviate some of this administrative burden. One review proposed a four-stage roadmap for expanding an ambient scribe's role in primary care workflows, from drafting clinical documentation (Stage 1), to drafting administrative artifacts such as letters/forms and follow-up tasks (Stage 2), to clinician-in-the-loop decision support during visits (Stage 3), and finally to proactive prompts that flag potential missed opportunities or concerns for clinician review (Stage 4)[23]. In Stage 1, these models automated essential clinical documentation, such as transcribing conversations with patients and generating watermarked, clinician-reviewable draft documentation (e.g., visit notes and structured summaries) that clinicians can verify, edit as needed, and attest to before being entered into the EHR. Because information entered into the EHR becomes part of the official medical record and may have clinical, regulatory, and legal consequences, automation should prioritize drafting and summarization with clinician verification, supported by provenance and audit trails and periodic quality review, rather than unattended record updates. This allows clinicians to devote more time to clinical reasoning and patient interaction by reducing clerical EHR work, for example, drafting a first-pass note for clinician review, populating structured fields, reconciling medication lists, summarizing prior results into the record, routing inbox tasks, and

completing forms, while preserving clinician responsibility for synthesizing the assessment and plan. Stage 2 built on this foundation by enhancing administrative workflows. LLMs assisted in drafting letters, completing forms, and creating follow-up tasks, which streamlines the everyday responsibilities that often consume time in primary care settings. In Stage 3, LLMs delivered reactive clinical decision support by addressing clinician inquiries during patient visits. For instance, they might offer medication dosage suggestions or propose possible diagnoses based on the patient's symptoms and medical background. Stage 4 presented the most advanced application, where the LLM assumed a proactive stance in clinical decision support. It alerted clinicians to missed opportunities or potential concerns throughout the visit. For instance, it could remind a clinician about recent lab results, like abnormal blood pressure[24] or high cholesterol levels, acting as a partner in patient care, actively aiding clinicians in their decision-making processes and enhancing the overall quality of care within the primary care environment[23].

Another area where LLMs have shown efficacy is with healthcare portal messaging and patient medical advice requests, where LLM-assisted drafting can improve perceived message quality (e.g., tone, clarity, and overall acceptability) when clinicians review and finalize the content. In one study looking at a certain LLM using prompt engineering, the percentage of physicians who found the messages suitable increased from 62% to 84%. Moreover, 74% of the drafts were considered "helpful." Patients also expressed a notably more positive perception of the message tone (78%) and overall quality (80%) when draft replies were generated using the optimized prompt, compared with draft replies generated using the baseline (non-optimized) prompt template. In fact, patients were unable to tell the difference between human and LLM-generated draft patient medical advice request responses for 76% of the messages, a shift from their earlier preference for human-generated responses. Most clinicians (72%) felt that the new prompts could help reduce the cognitive burden of managing EHR messages[25]. Another study comparing LLM-assisted EHR messaging versus clinician-only responses found similar results; the Comprehensive LLM Artificial Intelligence Responder (CLAIR): a locally fine-tuned LLM model, had the best performance by exhibiting higher clarity, conciseness, and utility than healthcare clinicians[26].

Diabetes. Within primary diabetes care, LLM-enabled systems, particularly when combined with deep learning (DL), have been evaluated for improving diabetic retinopathy identification/screening compared with PCPs alone. This is relevant to primary care because diabetes management in primary care includes coordinating complication screening and referral follow-through for conditions such as diabetic retinopathy. One prospective study showed that an LLM module demonstrated superior diagnostic performance on par with PCPs and even endocrinology residents[8]. When combined with DL technology, the LLM significantly enhanced the accuracy of diabetic retinopathy identification, elevating PCP accuracy from 81.0% to 92.3%[8]. These patients also showed improved self-management behaviors and a higher likelihood of adhering to diabetic retinopathy referrals compared to those managed by unassisted PCPs. Furthermore, the DL-LLM framework enhanced the quality and empathetic nature of management recommendations, providing a viable digital solution for advancing the screening of diabetic retinopathy[8].

Geriatric care. In outpatient primary care, "reception" and intake workflows include the front-desk and the nursing activities that collect the patient's presenting concern, clarify its urgency and the key symptoms, reconcile basics (i.e., medications, problems), and route the patient to the appropriate next step (i.e., schedule a visit, nurse assessment, clinician review, or escalation). These steps can be especially challenging in older adults, who often present with multimorbidity, polypharmacy, sensory and cognitive limitations, as well as complex narratives, leading to repeated questioning, delays, and frustration for patients and staff.

Against this backdrop, Wan et al.[5] evaluated an LLM-enabled chatbot (SSPEC) for outpatient reception. Patients assigned to the intervention primarily interacted directly with the SSPEC chatbot through an audio interface to ask administrative, triage, and primary care–related questions. When the system detected uncertainty in a response, it alerted a receptionist nurse to review the exchange and intervene as needed. Compared with nurse-only reception, this patient-facing, nurse-supervised workflow reduced repeated rounds of questioning and the frequency of negative emotional expressions by patients during the interaction. In a randomized controlled trial involving 2164 participants, the nurse–SSPEC group reported higher patient satisfaction and improved communication quality; repeated questioning occurred less frequently, and negative patient emotions were recorded in 2.4% of interactions compared with 7.8% in the nurse-only group. This innovative model showcased how LLMs can positively affect patient interactions, streamline healthcare processes, and enhance experiences for both patients and nurses, particularly in communication-centric settings like geriatrics[5].

Improving front-door patient intake and routing can reduce redundant processes, shorten the time-to-appropriate care, and support safer escalation pathways for older adults, outcomes that primary care practices can evaluate prospectively (e.g., time-to-triage decision, mis-triage or safety events, staff workload, and subgroup equity).

Cardiology. In primary care, an important and recurring challenge is closing the loop on incidental cardiovascular findings documented on imaging performed for another clinical indication[27]. Common examples of routine thoracic imaging include coronary artery calcification, aortic valve calcification, aortic root or thoracic aortic dilatation, cardiac chamber enlargement, and pericardial effusion. Depending on their severity and clinical context, these findings may prompt cardiovascular risk-factor assessment, additional imaging, surveillance, or specialist referral. Ensuring appropriate outreach and follow-up can be difficult when case identification depends on manual review of radiology reports.

In this context, one study evaluated an NLP approach to identify and extract incidentally detected aortic aneurysm findings and key measurements, such as aortic diameter, from radiology reports and compared its performance with traditional manual review. In the study cohort ($n = 412$), the tool enabled identification of incidental findings (58/412) and supported downstream outreach and follow-up tracking. Why this matters in primary care: this approach does not require primary care clinicians to "interpret imaging"; rather, it can reduce results-management burden and improve timeliness and reliability of follow-up workflows for incidentally detected aortic aneurysm findings, where primary care teams often coordinate outreach and referrals. Future primary care-relevant evaluations should measure follow-up completion and timeliness, false-positive/false-negative downstream consequences, workload impacts on care teams, and equity in successful contact and follow-up[27].

Radiology. Although primary care clinicians do not interpret MRIs or CTs, delays in radiology prioritization and reporting can affect primary care workflows (e.g., time-to-diagnosis, escalation decisions, and referral timeliness); we therefore include this as adjacent-setting, task-relevant evidence rather than primary-care-setting evidence.

A study designed to use a DL model to classify brain MRIs as normal or abnormal, specifically using axial Fluid-Attenuated Inversion Recovery (FLAIR) sequences, showed commendable performance, achieving F1 scores and AUROC metrics above 0.80. This implies that the model could effectively help in creating a prioritized worklist of cases that are likely to be abnormal in radiology, thus enhancing workflow efficiency. Accordingly, this type of radiology triage may help prioritize potentially abnormal studies for earlier radiologist review, which could shorten time to actionable results for referring clinicians (including primary care) and thereby support timelier clinical follow-up, though this downstream primary-care impact was not evaluated in the study. Overall, the results reveal that the model was able to differentiate between normal and abnormal brain scans with a good level of accuracy[28]. Another study looked at non-contrast head CTs (NCCTs), which have traditionally shown limited capability in detecting

early strokes (within 3–6 h). The DL model, trained on 3566 pairs of NCCT and MRI images, surpassed experienced neuro-radiologists with 150 CT scans. It achieved a sensitivity of 96% (with a specificity of 72%), whereas the experts exhibited a sensitivity range of 61–66% and a specificity of 90–92%. Moreover, the model's estimates for infarct volumes demonstrated a strong correlation with diffusion MRI results ($r^2 > 0.98$). When tested on a broader collection of 364 CT scans featuring a diverse array of infarct locations, the model's performance further improved, attaining a sensitivity of 97% and a specificity of 99% for detecting infarcts. These results highlight the considerable potential of AI in enhancing early stroke detection via NCCT, thereby aiding radiologists in delivering quicker and more precise diagnoses[28].

Future evaluations should measure primary care-relevant endpoints such as time-to-final report, time-to-PCP action, follow-up completion, and unintended consequences (i.e., false-positive cascades, missed abnormal results), including subgroup equity.

Neurology. Neurology and neurosurgical admissions frequently transition back to primary care, where clinicians are responsible for medication reconciliation, monitoring for complications, and coordinating follow-up[29]. In this context, discharge summaries serve as a key communication item for PCPs who are downstream "consumers" of the document. A study evaluating 60 neurosurgical discharge summaries reported that ChatGPT substantially reduced drafting time compared with a conventional speech-recognition workflow (SpeaKING), with median times decreasing from 15–21 min to 2.3–4.6 min across representative neurosurgical scenarios (chronic subdural hematoma, one-level spinal decompression, and craniotomy for tumor resection)[29]. While these findings support potential efficiency gains in documentation workflows, the study did not formally assess whether LLM-generated discharge summaries were more clinically actionable for primary care (e.g., clarity of medication changes, pending results, red-flag symptoms, and follow-up plans). Future evaluations should explicitly incorporate primary care-facing usability and downstream care-coordination outcomes. In practice, all discharge summaries, including those drafted with AI assistance, should use a standardized structure that clearly identifies medication changes, pending test results, and responsible clinicians, follow-up plans, contingency instructions, and warning symptoms relevant to patients and primary care teams.

Behavioral health triage and routing. Although relatively few studies have evaluated LLM-supported triage within primary care clinics, several investigations in adjacent frontline settings illustrate how language-model-based or NLP approaches can be applied to classification, urgency stratification, and routing tasks that are also central to primary care. In emergency care, an admission-prediction model built from ED triage notes predicted patient disposition (admission vs non-admission) and reported 83% accuracy, with recall (sensitivity) 72% and precision (positive predictive value) 56% in the original (offline) evaluation[30], and a subsequent real-time post-implementation study reported 74% overall accuracy, with the lowest accuracy for psychiatric-unit admissions (34%), underscoring challenges in mental health presentations[31].

Related ED work has also led to supervised NLP classifiers aiming to detect self-harm/suicidal ideation directly from triage notes, achieving approximately 90% precision and 90% recall on blind test data with similar prospective validation[32]. Separately, psychiatry-focused work in specialist contexts has evaluated machine-learning triage of psychiatric cases (e.g., severity stratification across four classes from textual psychiatric records), achieving 77% overall accuracy in one cascading-classifier approach[33]. However, these settings are not interchangeable with primary care: differences in case-mix, available diagnostics, staffing, time horizons, and escalation options mean that triage performance observed in emergency departments or specialist services should not be assumed to generalize to primary care without direct evaluation. Future studies should therefore test triage and routing support in primary care-specific pathways (e.g., telephone/nurse triage, portal message routing, and referral management) using downstream outcomes such as safety events, appropriateness of escalation, timeliness of care, and subgroup equity.

Medical knowledge assessment and educational support. Evidence frequently cited to contextualize LLM capabilities comes from medical education benchmarks, including performance on licensing-style examinations. These results are best interpreted as an indirect proxy for breadth of medical knowledge and text-based reasoning rather than as evidence that LLMs assist clinicians in completing primary care tasks. For example, in a UK observational benchmark using the MRCGP Applied Knowledge Test, ChatGPT-3.5 performed below the typical pass threshold and showed variability across topic areas, illustrating both capability and clinically relevant gaps[34].

In primary care, the most relevant implication of such benchmarks is the potential for LLMs to support just-in-time assisting (e.g., clarifying guideline concepts, summarizing differential diagnoses, or drafting patient-facing explanations) when used with appropriate oversight and grounding in trusted references34. Importantly, strong performance on knowledge assessment does not establish safe or effective clinical use: real-world primary care assistance requires evaluation in workflows that involve uncertainty, incomplete information, time pressure, accountability, and patient-specific context.

Accordingly, these benchmarks are most relevant for establishing educational frameworks and informational support (e.g., clarifying guideline concepts or drafting patient-facing explanations) and should not be interpreted as evidence that LLMs can replicate the broader expertise and accountability of a qualified primary care clinician.

Specialist-informed decision support in primary care: escalation and referral. LLM-enabled tools may support primary care clinicians by providing specialist-informed decision support, for example, helping identify when escalation is warranted, improving urgency stratification, and generating higher-quality referral justification, particularly in settings where timely specialist access is limited. A recent study assessing an LLM's ability to answer 30 multiple-choice questions related to primary healthcare revealed an impressive accuracy rate of 93.33%. The model also scored well in terms of explanation quality, averaging 4.58 ± 0.85, and showed remarkable agreement among reviewers (ICC of 0.94, 95%). These findings emphasize the potential usefulness of LLMs in supporting PCPs, especially in rural or underserved regions where specialist access may be scarce[35].

It is conceivable that LLM-enabled tools could, in some scenarios, broaden access to decision support in settings with limited specialist availability. However, evidence that solely providing LLM access improves clinicians' diagnostic reasoning beyond conventional resources remains limited. In a randomized clinical trial, physician diagnostic-reasoning performance with access to an LLM did not significantly differ from performance using conventional diagnostic resources alone[4,17]. With their ability to ingest complex narrative and structured clinical text and generate human-readable outputs, LLMs may help clinicians rapidly summarize histories, highlight salient findings, and draft candidate differentials or management considerations for clinician review. In vignette-based evaluations, aggregating outputs across multiple LLMs improved differential-diagnosis accuracy compared with single models[8], and LLMs have also shown high performance on primary-care question sets, supporting potential roles as educational frameworks and remote health-information support rather than demonstrated substitution for specialty consultation[29]. Importantly, these studies did not evaluate rural vs urban care settings or primary care clinic workflows; prospective, workflow-based evaluation is needed before inferring impact on access, referral patterns, or outcomes.

In interventional cardiology, a proof-of-concept study evaluated commercially available LLMs in simulated multidisciplinary heart-team decisions concerning structural and coronary interventions. (e.g., TAVR vs. SAVR; PCI vs. CABG)[36]. Because these decisions occur within specialist care

after referral, the study does not provide direct evidence for LLM use in primary care or for improving referral decisions made by primary care clinicians. We include it only as adjacent-setting evidence illustrating the current capabilities and limitations of LLM-supported clinical decision-making in a highly specialized context. Its findings should not be extrapolated to primary care workflows or interpreted as evidence that LLMs can replace specialist expertise. For primary care, the more relevant questions are whether AI tools can help identify when referral is warranted, determine urgency, improve the completeness of referral information, and support appropriate follow-up outcomes that require direct prospective evaluation in primary care settings.

In parallel, real-world deployment of non-LLM AI tools in Australian primary care has demonstrated feasibility, accuracy, and acceptability for automated retinal photography and AI-based cardiovascular risk assessment, illustrating how decision support can be operationalized in primary care workflows[37].

Population health and prevention in primary care. Much of the public health promise of LLMs in primary care remains hypothesis-generating; evidence is strongest for workflow augmentation and weakest for demonstrated population-level outcome improvements.

The future integration of LLMs into primary care holds transformative potential for addressing prevalent public health challenges and enhancing clinical workflows. As primary care physicians increasingly manage patients with complex chronic conditions such as diabetes, cardiovascular disease, and geriatric multi-morbidity (including polypharmacy), LLM-enabled tools may support clinician review by synthesizing structured and unstructured EHR data to surface salient findings, stratify risk, and draft evidence-aligned diagnostic or management considerations. By synthesizing structured and unstructured data (e.g., EHRs, clinical notes, and diagnostic results), LLM-enabled tools may help bring to the surface under-recognized risk signals, early complications, and care gaps (e.g., missed follow-up, worsening control, or polypharmacy-related hazards) and support risk stratification and drafting of evidence-aligned management considerations for clinician review. This is likely to be more clinically meaningful than simply detecting the presence of common conditions (e.g., diabetes or obesity) that primary care clinicians can usually identify, and instead pertains to identifying patients at risk of complications or needing timely follow-up[38,39].

Moreover, LLM-enabled tools may support preventive care by helping clinicians and care teams synthesize patient information and draft guideline-concordant education, counseling, and care-plan elements for clinician review. More generally, retrieval-based grounding approaches (e.g., retrieval-augmented generation) can bind model outputs to an updatable external evidence base, such as clinical guidelines[40]. In diabetes care, Sheng et al. note that a key mitigation strategy for outdated or misaligned outputs is to adapt models with domain-specific resources (e.g., clinical practice guidelines and specialty knowledge) to improve the consistency of generated recommendations[41]. Such approaches could enable primary care settings to more proactively address the needs of at-risk populations, supporting earlier intervention and reducing downstream burden pending prospective evaluation in real workflows.

In addition, as the accessibility of AI technologies improves globally, LLMs could play a critical role in addressing health disparities in underserved regions. By offering decision-support tools in multiple languages and cultural contexts, these models can assist clinicians in providing equitable care, particularly in areas with limited healthcare resources[26,42]. For instance, multilingual LLMs can facilitate patient education, explain complex medical conditions in culturally sensitive terms, and assist with medication adherence in populations facing language barriers[43].

Cost-minimizing by using LLMs in primary care

Early evidence suggests LLM-enabled tools may be cost-effective in primary care primarily by reclaiming clinician time and reducing avoidable, low-yield work and use of service resources (e.g., duplicative documentation, inefficient triage that generates unnecessary visits/tests, and administrative burden), especially when embedded within established digital-first pathways and supported by interoperable data exchange[44,45]

As discussed in the Electronic Health Records section, ambient "AI scribe" deployments at a large U.S. multispecialty group reported substantial documentation time savings, equivalent to ~1800 physician-days in a year, alongside positive clinician and patient experience, implying meaningful labor-cost offsets without adding visits[46]. Because primary care teams handle high volumes of portal messages, even small per-message reductions in clinician effort (or improved draft acceptance that reduces revision time) can aggregate into meaningful operational savings at practice or health-systems, although measured time savings have been mixed and should be evaluated prospectively[47].

At the infrastructure level, a health-system-wide study in England found that primary-to-secondary care data sharing improved emergency flow and patient experience; this study underscores the interoperability as a prerequisite for LLM-enabled workflow automation to access the necessary clinical context and reduce revisions and delays[46]. Complementing this, real-world "digital-first" primary-care models (asynchronous triage/chat with escalation) delivered 22.7% lower episode costs on average across five common minor conditions, mainly by trimming encounter and test costs; LLM-enabled tools could potentially amplify these gains by automating history-taking, triage, and documentation within the same pathways.

From a practical cost perspective, the expense of running LLMs can be reduced by using smaller, lower-cost models for routine tasks (e.g., drafting standard portal replies) and reserving larger models for complex cases, and by processing high-volume work more efficiently at scale; such strategies have been shown to substantially lower operating costs in health-system settings, strengthening the feasibility of their routine use[48]. Altogether, the emerging picture is that LLMs are most cost-effective when (1) they perform clerical work (notes, messages) rather than adding steps; (2) they are embedded in digital-first workflows already shown to lower costs; and (3) they ride on interoperable data streams that minimize rework and delays[46,49]. However, important caveats remain; benefits vary by task (e.g., inbox time unchanged in some studies), rigorous prospective cost-outcome evaluations are still sparse, and governance/safety standards must keep pace—but the direction is toward net savings with maintained quality when implemented thoughtfully[12,47,50,51].

LLM governance and policy considerations for primary care deployment

Despite their potential, the use of LLMs in primary care is not without challenges, particularly around patient data privacy, security, and ethical policy concerns (Table 1)[52–54]. One of the most pressing issues is ensuring compliance with the Health Insurance Portability and Accountability Act (HIPAA), which mandates strict standards for the protection of patients' personal health information. Any system that processes or stores health data, including LLMs, must be designed to ensure that personal health information remains confidential and is only accessible to authorized individuals. This includes implementing strong encryption protocols, regular audits, and access controls to prevent unauthorized data access[55]. In the same vein, the EU AI Act has implemented a risk-based framework that categorizes numerous healthcare AI applications, such as LLM diagnostic tools and clinical decision support systems, as "high-risk."[56] In the United States, the FDA has provided guidance that outlines the expectations for the safe development, validation, and modification of AI-based medical devices[57], however, it has not yet approved an LLM-enabled medical device. Additionally, the ONC HTI-1 (Health Data, Technology and Interoperability) final rule reinforces transparency requirements for AI algorithms[44,45]. These initiatives signal a transition toward a more structured, interoperable, and reliable use of AI in healthcare[44,45].

Ethical and equity considerations

Furthermore, LLMs trained on medical data must be carefully monitored for biases that could lead to inequitable care, such as algorithms that might

over- or under-represent certain demographic groups[58], potentially exacerbating health disparities (Table 1)[59], negatively impacting health-care delivery. There is also the concern that LLMs, if not appropriately trained or supervised, could generate incorrect or misleading information, which could impact clinical decision-making[60]. Beyond privacy, cybersecurity threats are a significant concern. LLMs are often cloud-based systems, meaning they are susceptible to cyberattacks that could compromise patient data. Cybersecurity protocols, such as multi-factor authentication, regular penetration testing, and rapid response systems for potential breaches, must be in place to safeguard sensitive information. The intersection of these technical and ethical considerations highlights the need for rigorous safeguards and thoughtful implementation when integrating LLMs into clinical workflows (Table 1)[61,62].

Conclusions

In primary care settings, LLMs bring numerous advantages that can elevate patient care and refine healthcare workflows (Fig. 1)[63–65]. As previously mentioned, these models have helped enhance patient-clinician communication and alleviate the cognitive load on healthcare providers, thereby influencing practice. They also have aided physicians with administrative tasks such as drafting discharge summaries, transcribing patient conversations, and managing messages through healthcare portals, affecting clinical workflows.

In the current evidence base, the most consistently supported near-term value of LLMs in primary care is workflow augmentation, particularly in documentation and inbox management (e.g., drafting portal replies and summarizing information for clinician review), which can improve perceived communication quality and reduce cognitive burden when clinicians retain responsibility for verification and final decisions (Fig. 1). Evidence for improvements in clinician diagnostic reasoning and downstream patient outcomes is more limited and context-dependent, and many studies remain simulated or conducted in adjacent settings; accordingly, claims of improved patient outcomes should be currently treated as hypotheses that require prospective evaluation. We therefore emphasize a pragmatic implementation approach: prioritize LLM utilization on high-volume, low-risk clerical and communication workflows; apply safeguards and governance (Table 1); and evaluate impact using primary care relevant endpoints (e.g., time, safety events, follow-up completion, and equity) before scale-up.

LLMs may also support hospital-linked and integrated-care pathways relevant to primary care, including transitions of care, by synthesizing clinical information and generating clinician-reviewable summaries or decision-support considerations under appropriate supervision[66].

Managing rapidly evolving clinical information (including during public health emergencies) underscores the importance of clear governance for any generative AI used in clinical care. The Society of General Internal Medicine position statement provides recommendations for clinicians, technologists, and healthcare organizations on responsible deployment of generative AI (including oversight, transparency, and organizational policies) and emphasizes that these tools should augment, and not replace, clinical judgment[67]. While the widespread adoption of LLMs in primary care will require overcoming challenges such as data privacy concerns[53] and ensuring compliance with regulations like HIPAA, their potential to revolutionize diagnosis, treatment, and preventive care underscores their role in the future of healthcare delivery. In Table 1, we offer recommendations pertinent to the use of LLMs in primary care, such that by strategically addressing these limitations, primary care can fully harness the capabilities of LLMs to improve patient outcomes and streamline care in an increasingly complex healthcare landscape (Fig. 1).

Primary care professional organizations and academic groups have begun curating practical guidance for responsible AI adoption. The American Academy of Family Physicians (AAFP) maintains an "AI in family medicine" resource hub emphasizing education, research, and reducing administrative burden[68]. In the Journal of the American Board of Family Medicine, Hanna et al. outline why family medicine should proactively prepare for AI integration and highlight implications for practice, training, and governance[69]. In parallel, the Stanford Healthcare AI program curates primary care–relevant AI publications and perspectives, providing a navigable entry point into a rapidly evolving evidence base[70]. These resources reinforce that primary care needs workflow-based evaluation, human oversight, and organizational policies that define acceptable use, documentation standards, and monitoring expectations.

Limitations

Because primary care use-cases are predominantly text- and workflow-mediated (e.g., documentation/inbox, portal messaging, triage and routing, as well as clinician decision support), we summarize risks and safeguards in terms of how they manifest in these specific workflows. Table 1 provides a practical checklist for implementation. The evidence base summarized here is heterogeneous, with study designs and analytic approaches spanning randomized trials, retrospective evaluations, prospective cohorts, and cross-sectional surveys. Many reports rely on simulated tasks, single-center data, short follow-up, and proprietary model configurations, which limit generalizability and increase susceptibility to selection, measurement, and confounding biases[6,7]. Accordingly, conclusions should be interpreted as practice-relevant and hypothesis-generating, and further pragmatic studies are needed to quantify real-world effects on patient outcomes, safety, equity, and cost (Fig. 1).

Data availability

No new data were generated for this manuscript. All studies referenced are included in the Main reference list.

Received: 17 March 2026; Accepted: 20 July 2026;
Published online: 25 August 2026

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Acknowledgements

We sincerely thank the anonymous reviewers for their careful reading of the manuscript and for providing constructive, detailed feedback. Their insights substantially improved the clarity, organization, and scope of the review, strengthened the primary care framing, and helped us refine the discussion of evidence limitations, implementation safeguards, and policy-relevant considerations.

Author contribution

M.C. authored and conceptualized the manuscript. K.P. authored the manuscript. J.Z. authored the manuscript. A.A.A. conceived, conceptualized, and authored the manuscript

Funding

A.A.A. is funded by the Institute of Precision Medicine (17UNPG33840017) of the American Heart Association, the RICBAC Foundation, NIH grants 1 R01 HL135335-01, 1 R01 HL161008-01, 1 R21 HL137870-01, 1 R21EB026164-01, and 3R21EB026164-02S1.

Competing interests

The authors declare no competing interests

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43856-026-01823-z.

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Peer review information *Communications Medicine* thanks Pushpa Kumarapeli, John D McGreeveyand the other anonymous reviewer(s) for their contribution to the peer review of this work.

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